# SARATHKUMAR DUNNAPAPAGARI

M.Tech CSE Student · B.Tech IT Graduate · Fresher

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LinkedIn GitHub Portfolio

## PROFILE SUMMARY

Motivated M.Tech Computer Science student and B.Tech IT graduate with four internships and trainings in AWS cloud, Python development and data science / machine learning. Hands-on experience building real-time computer vision, machine learning and full-stack projects with Python, PyTorch, YOLOv8, FastAPI and React, and deploying on AWS and GCP. Eager to start my career in a full-time role where I can learn fast and build reliable, scalable solutions.

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages** | Python · Java · C |
| **Cloud** | AWS EC2 · S3 · IAM · Google Cloud (GCP) |
| **ML & Data** | PyTorch · scikit-learn · YOLOv8 · OpenCV · Pandas · DuckDB |
| **Web & APIs** | FastAPI · React · HTML · CSS · JavaScript |
| **Tools** | Docker · Git & GitHub · Prometheus · Grafana · Linux · MS Office · AutoCAD |

## PROJECTS

**Construction Site Safety Monitor** GitHub · Project page — **Computer Vision · Cloud**

- Built a real-time YOLOv8 pipeline for webcam, CCTV or video files that streams live detections to a React dashboard.
- Designed PPE-ready alerts: helmet / vest violations with S3 snapshots switch on once a custom PPE model is configured.
- Containerized with Docker Compose and NGINX, with an EC2 setup script and Prometheus + Grafana monitoring.

**Stack:** Python · YOLOv8 · OpenCV · FastAPI · React + TypeScript · Docker · AWS S3 · Prometheus · Grafana

**Data Analyst + ML Agent** GitHub — **Full-stack · ML**

- Developed an AI-assisted data analysis platform: explore datasets with charts, plain-English questions or SQL, then train and compare ML models.
- Added PDF reports and optional LLM explanations on code-computed results; works fully without an API key.

**Stack:** FastAPI · Pandas · DuckDB · scikit-learn · React · TypeScript · Tailwind CSS

**EmoMusic** GitHub — **Deep Learning**

- Created a music recommender driven by real-time facial emotion recognition that eases the listener toward a calmer mood.
- Fine-tuned a transfer-learned CNN on FER-2013 and smoothed predictions so the mood and playlist stay stable.

**Stack:** PyTorch · timm · OpenCV

## EDUCATION

**M.Tech, Computer Science and Engineering** GMR Institute of Technology, Rajam *Pursuing* — **2025 – 2027**

**B.Tech, Information Technology** Sir C.R. Reddy College of Engineering, Eluru **CGPA 7.29** — **2021 – 2024**

**Diploma, Civil Engineering** M.R.A.G.R Government Polytechnic, Vizianagaram **67.69%** — **2017 – 2020**

## INTERNSHIPS

**AWS Cloud Intern** BrainOVision Solutions — **2 months**

- Deployed apps on AWS EC2, managed S3 storage, explored IAM roles, and set up and monitored cloud infrastructure.

**Data Science & Machine Learning Intern** APSCHE — **6 months**

- Worked on predictive modeling, applying ML algorithms and improving accuracy through data preprocessing.

**Python Developer Intern** BrainOVision Solutions — **2 months**

- Developed Python automation scripts and small applications, improving efficiency with optimized code.

**Industrial Training** APSPHC — **6 months**

- Gained exposure to infrastructure and technical workflows in public sector projects.

## CERTIFICATIONS

- ✓ **Google Cloud Computing Foundations** NPTEL
- ✓ **Cloud Computing** NPTEL
- ✓ **Introduction to Soft Skills** TCS
- ✓ **Getting Started with Amazon Web Services** SkillUp
- ✓ **FDP on Cloud Infrastructure** BrainOVision